

SEAT No. _____

[49]

SARDAR PATEL UNIVERSITY
B.Sc. (CA&IT) EXAMINATION - JULY 2021
6TH Semester (CBCS)(REGULAR)
Friday, Date: 23/07/2021

Session : Evening Time : 02:00 P.M. to 04:00 P.M.

Subject Code: PS06CIIT23

Subject Title : Computer Networks

Total Marks: 70

Q1. Multiple Choice Questions

(10)

1. _____ network consists of many connections between individual parts of machines.
 - a. Specialized
 - b. Multicast
 - c. Point-to-point
 - d. Broadcast
2. If a computer on the network allows shared resources for others to use, it is called _____.
 - a. Router
 - b. Client
 - c. Server
 - d. Host
3. _____ is a digital multiplexing technique for combining several low-rate channels into one high-rate one.
 - a. FDM
 - b. TDM
 - c. WDM
 - d. CDMA
4. _____ is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.
 - a. CDMA
 - b. Demultiplexing
 - c. Serial transmission
 - d. Multiplexing
5. A _____ cable consists of two conductors each with its own plastic insulation, twisted together.
 - a. Twisted pair
 - b. Coaxial
 - c. Fiber Optics
 - d. None of the above
6. An artificial satellite needs to have _____ which is the path in which it travels around the Earth.
 - a. Channel
 - b. Footprint
 - c. Tower
 - d. Orbit
7. Bit synchronization is responsibility of _____ layer.
 - a. Application
 - b. Physical
 - c. Presentation
 - d. Session
8. Encryption and compression is responsibility of _____ layer.
 - a. Presentation
 - b. Transport
 - c. Network
 - d. Application
9. _____ network device can be used to solve problem of attenuation.
 - a. Amplifier
 - b. Repeater
 - c. Hub
 - d. Switch
10. A router operates at _____ layer.
 - a. Physical
 - b. Application
 - c. Session
 - d. Network

Q2. Fill in the blanks and True - False**(08)**

1. "Interconnected collection of autonomous computers" is known as ____.
2. In a fiber-optic cable, the signal is propagated along the inner core by ____.
3. Mail and directory service is responsibility of ____.
4. A switch performs filtering on the basis of ____ address.
5. Subnet consists of two distinct components: transmission lines & switching elements. (True/False)
6. TDM is an analog multiplexing technique that combines analog signals. (True/False)
7. A set of layers and protocols is called Computer Architecture. (True/False)
8. A router forwards packets on the basis of their logical address. (True/False)

Q3. Short Answer Questions (Attempt any 10 out of 12)**(20)**

1. Explain DQDB in brief.
2. Explain Broadcast network in short.
3. Define the Terms : (i) Topology (ii) Data rate
4. Explain Amplitude modulation in brief with example.
5. Explain Parallel transmission in short.
6. List various categories of Coaxial cable and state their use.
7. Define: Protocol Stack and Network architecture.
8. Explain Footprint.
9. What is Satellite? List the categories of Satellite.
10. Explain CSMA/CD in short.
11. List out advantages and disadvantages of Star topology.
12. Explain Bluetooth in short.

Q4. Long Answer Questions (Attempt any 4 out of 8)**(32)**

1. Explain the Advantages and Disadvantages of computer networks in detail.
2. Explain LAN and WAN in Detail.
3. Explain any two types of Transmission media in detail.
4. Explain Synchronous TDM in detail.
5. Explain various layers of OSI model in detail.
6. Write a note on design issues for the layers.
7. Explain Bus, Mesh and Ring topologies in detail with its advantages and disadvantages.
8. Explain any 3 Network devices in detail.